or houses — or even light, magnetism, temperature — or anything the tricorder can help them see.”

The open twist: Dr Jansen’s tricorder project is completely open source. People can link it up with other systems and applications to access much more information, do deeper analysis, and use it for learning, experimentation or product development. The latest complete version is the Science Tricorder Mark 2. It runs Linux, and has a number of connectivity and development options. It comprises a suite of atmospheric, electromagnetic and spatial sensors. The upgradeable and self-contained sensor boards have separate processors and use an open sensor protocol. Other tech specifications include: Atmel AT91RM9200 (ARM920T 32-bit RISC core/ 180 MHz) processor, dual 7.11 cm (2.8") OLED touch displays, Debian Linux, USB host/device and SD card storage. While Mark 3 was abandoned mid-way, Mark 4 is now getting ready – it has been fabricated already and software development is in progress. The goal for the Mark 4 is to bring down the cost and make it more accessible to all.

Do-it-yourself virtual reality projects

http://projects.ict.usc.edu/mxr/diy/

There is a huge gap between the excitement generated around virtual reality and the actual resources available for those who want to use the technology in various projects. There are a lot of commercial products (especially headsets) available these days, but not much in terms of design, code, etc. In an effort to bridge this gap, the University of Southern California’s MxR Lab has launched a website that showcases headsets and modifications that DIY enthusiasts can build, including open source code for the devices and full-body motion control integration through Kinect for Windows or OpenNI.

The open twist: The website at present features two major headset designs and a developer kit, plus a lot of middleware, scripts, code and modifications. The Socket HMD is a complete 1280 x 800 headset with a 3D-printed shell and custom-assembled electronics. The VR2GO viewer is a simpler project, which uses iPhones or iPod touch players as the eye pieces, and mods from the Oculus Rift developer kit for stereo cameras. You may tweak the designs and come up with a viewer that best suits your need. However, you will need access to a 3D printer to get it printed!

Small and handy submarines, for the explorer in you

http://openrov.com/

OpenROV is a project that aims to open up the ocean to not just the adventurous but also hobbyists and educators. The project is led by NASA engineer Eric Stackpole, and promotes an open source underwater DIY robot. The underwater robot is capable of sinking to depths of up to 100 metres (although it has been tested only for 25m till date), runs on eight C-cell batteries for approximately 1.5 hours and has a speed of 1 m/s. The structure is made of a sealed cylinder within a laser-cut acrylic frame. The cylinder houses a BeagleBone, high-definition (HD) webcam and LED lights. It is tethered to a laptop on the shore by a single twisted pair of cables communicating 10 megabit Ethernet data for control and video. The robot can be piloted using a Web browser and video feed.

The open twist: You could either buy a complete kit from the website to build an OpenROV, or use the designs and list of materials needed to source whatever you need from a local electronics store. The design is organised into various sections, most of which can be readied within